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Section :- 5A

Course :- Electromagnetic Theory.

Q#01

$$D = 4xy \hat{a}_x + 2(x^2 + z^2) \hat{a}_y + 4yz \hat{a}_z \quad \text{nC/m}^2$$

Evaluate surface integral to find total charge enclosed in rectangular parallelepiped

$$0 < x < 2, 0 < y < 3, 0 < z < 5 \text{ m.}$$

Solution:-

$$\rho_v = \nabla \cdot D$$

$$\rho_v = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\frac{\partial D_x}{\partial x} = \frac{\partial (4xy)}{\partial x} = 4y.$$

$$\frac{\partial D_y}{\partial y} = \frac{\partial (2(x^2 + z^2))}{\partial y} = 0$$

$$\frac{\partial D_z}{\partial z} = \frac{\partial (4yz)}{\partial z} = 4y$$

$$\rho_v = 4y + 0 + 4y$$

$$\rho_v = 8y \quad \text{nC/m}^2$$

→ For Total Charge

$$Q = \oint D \cdot ds$$

$$Q = \int p_v dv$$

$$Q = \int_0^2 \int_0^3 \int_0^5 8y \, dz \, dy \, dx$$

$$Q = \int_0^2 \int_0^3 8yz \Big|_0^5 \, dy \, dx$$

$$Q = \int_0^2 \int_0^3 40y \, dy \, dx$$

$$Q = \int_0^2 \frac{40y^2}{2} \Big|_0^3 \, dx$$

$$Q = \int_0^2 20(3-0)^2 \, dx$$

$$Q = 180x \Big|_0^2$$

$$Q = 360 \, \text{nC/m}^2$$

Q #02

Volume charge of constant density $p_v = p_0$
 $-\infty < x < \infty$, $-\infty < y < \infty$, $-d/2 < z < d/2$.

Find D and E everywhere.

Solution:-

From the statement, the field is in z -direction and also uniformly with x & y leaving z -field. Thus, the Gaussian surface will be enclosed in a rectangular region is given

$$-1 < x < 1, \quad -1 < y < 1, \quad |z| < \frac{d}{2}$$

The outward flux from this surface will be limited at $\pm z$.

Now, we know that:-

$$\begin{aligned}\Phi_{in} &= \oint \mathbf{D} \cdot d\mathbf{s} = Q_{enclosed} \\ &= \int_{-z}^z \int_{-1}^1 \int_{-1}^1 \rho_v dx dy dz \\ &= \int_{-z}^z \int_{-1}^1 \int_{-1}^1 \rho_0 dx dy dz\end{aligned}$$

Both D_z and ρ_0 are constant.

$$D_{in} = \rho_0 z \hat{a}_z \quad \text{C/m}^2 \quad (|z| < d/2)$$

$$\text{Now, } E_{in} = (\rho_0 z / \epsilon_0) \hat{a}_z \quad \text{V/m} \quad (|z| < d/2)$$

So,

$$\begin{bmatrix} D_{in} = \rho_0 z \hat{a}_z & \text{C/m}^2 \\ E_{in} = \rho_0 z / \epsilon_0 \hat{a}_z & \text{V/m} \end{bmatrix}$$

→ For Outside:-

Gaussian surface will be the same, except parallel boundaries at $\pm z$ occurs at $|z| < d/2$

Now,

$$\begin{aligned}\Phi_{outside} &= \oint \mathbf{D} \cdot d\mathbf{s} \\ Q_{enclosed} &= z \int_{-1}^1 \int_{-1}^1 D_z dx dy \\ &= \int_{-d/2}^{d/2} \int_{-1}^1 \int_{-1}^1 \rho_0 dx dy dz\end{aligned}$$

We get

$$D_{out} = \begin{cases} (\rho_0 d/2) \hat{a}_z & (z > d/2) \text{ C/m}^2 \\ -(\rho_0 d/2) \hat{a}_z & (z < d/2) \text{ C/m}^2 \end{cases}$$

$$E_{out} = \begin{cases} (\rho_0 d/2 \epsilon_0) \hat{a}_z & (z > d/2) \text{ V/m} \\ -(\rho_0 d/2 \epsilon_0) \hat{a}_z & (z < d/2) \text{ V/m} \end{cases}$$

Q#03

$\nabla \cdot D$ if (a) $D = \left(\frac{1}{z^2}\right) [10xyz \hat{a}_x + 5x^2z \hat{a}_y + (2z^3 - 5x^2y) \hat{a}_z]$
at $P(-2, 3, 5)$

Solution:-

$$(a) \quad D = \frac{1}{z^2} [10xyz \hat{a}_x + 5x^2z \hat{a}_y + (2z^3 - 5x^2y) \hat{a}_z]$$

$$D = \left(\frac{10xy}{z}\right) \hat{a}_x + \frac{5x^2}{z} \hat{a}_y + \left(2z - \frac{5x^2y}{z^2}\right) \hat{a}_z$$

$$\nabla \cdot D = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

$$\frac{\partial D_x}{\partial x} = \frac{\partial \left(\frac{10xy}{z}\right)}{\partial x} = \frac{10y}{z}$$

$$\frac{\partial D_y}{\partial y} = \frac{\partial \left(\frac{5x^2}{z}\right)}{\partial y} = 0$$

$$\frac{\partial D_z}{\partial z} = \frac{\partial \left(2z - \frac{5x^2y}{z^2}\right)}{\partial z} = 2 - \frac{5x^2y(-2)}{z^3}$$

$$P(-2, 3, 5)$$

$$\nabla D = \frac{10y}{z} + \left(2 - \frac{5x^2y}{z^3} \right) \Big|_{-2, 3, 5}$$

$$\nabla D = \frac{(10)(3)}{(5)} + (2) + \frac{10(-2)^2(3)}{(5)^3}$$

$$\nabla D = 6 + \left(2 + \frac{120}{125} \right)$$

$$\nabla D = 8.96$$

$$(b) D = 5z^2 a_p^{\wedge} + 10p z a_z^{\wedge} \quad P(3, -45^\circ, 5)$$

$$\nabla D = \frac{1}{p} \left[\frac{\partial}{\partial p} (p D_p) + \frac{\partial}{\partial \phi} (D \phi) + \frac{\partial}{\partial z} (p D_z) \right]$$

$$\nabla D = \frac{1}{p} \left[\frac{\partial}{\partial p} (p(5z^2)) + \frac{\partial}{\partial \phi} (0) + \frac{\partial}{\partial z} (p(10pz)) \right]$$

$$\nabla D = \frac{1}{p} \left[(5z^2) + (0) + (10p^2) \right]$$

$$\nabla D = \frac{5z^2}{p} + 10p \Big|_{\substack{3, -45^\circ, 5 \\ -2, 3, 5}}$$

$$\nabla D = \frac{5(5)^2}{(3)} + (10)(3)$$

$$\nabla D = 71.66$$

(c) $D = 2r \sin \theta \sin \phi \hat{a}_r + r \cos \theta \sin \phi \hat{a}_\theta + r \cos \phi \hat{a}_\phi$
 $P(3, 45^\circ, -45^\circ)$

$$\nabla D = \frac{1}{r^2 \sin \theta} \left[\frac{\partial (r^2 \sin \theta D_r)}{\partial r} + \frac{\partial (r \sin \theta D_\theta)}{\partial \theta} + \frac{\partial (r D_\phi)}{\partial \phi} \right]$$

Plugging the values :-

$$\nabla D = \frac{1}{r^2 \sin \theta} \left[\frac{\partial (r^2 \sin \theta \cdot 2r \sin \theta \sin \phi)}{\partial r} + \frac{\partial (r \sin \theta \cdot r \cos \theta \sin \phi)}{\partial \theta} + \frac{\partial (r \cos \phi)}{\partial \phi} \right]$$

$$\nabla D = \frac{1}{r^2 \sin \theta} \left[\frac{\partial (2r^3 \sin^2 \theta \sin \phi)}{\partial r} + \frac{\partial (r^2 \sin \theta \cos \theta \sin \phi)}{\partial \theta} + \frac{\partial (r^2 \cos \phi)}{\partial \phi} \right]$$

$$\nabla D = \frac{1}{r^2 \sin \theta} \left[6r^2 \sin^2 \theta \sin \phi + r^2 \cos 2\theta \sin \phi - r^2 \sin \phi \right]$$

$$\nabla D = 6 \sin \theta \sin \phi + \frac{\cos 2\theta \sin \phi}{\sin \theta} - \frac{\sin \phi}{\sin \theta}$$

$$P(3, 45^\circ, -45^\circ)$$

$$\nabla D = 6 \sin(45^\circ) \sin(-45^\circ) + \frac{\cos 2(45^\circ) \sin(-45^\circ)}{\sin(45^\circ)} - \frac{\sin(-45^\circ)}{\sin(45^\circ)}$$

$$\nabla D = 2$$

Q#04

$$D = p_0(z+2d) \hat{a}_z \text{ C/m}^2$$

$$(-2d \leq z \leq 0) - p_0(z-2d) \hat{a}_z \text{ C/m}^2 \quad (0 \leq z \leq 2d)$$

Solution:-

(a) $\nabla D = p_v$

From statement

i. $D = p_0(z+2d) \hat{a}_z$ for $-2d \leq z \leq 0$

ii. $D = -p_0(z-2d) \hat{a}_z$ for $0 \leq z \leq 2d$

iii. $D = 0$ elsewhere

(i) $-2d \leq z \leq 0$

$$\nabla D = \frac{\partial}{\partial z} (p_0(z+2d))$$

$$\nabla D = p_0$$

(ii) $0 \leq z \leq 2d$

$$\nabla D = \frac{\partial}{\partial z} (-p_0(z-2d)) = -p_0$$

(iii) $\nabla D = 0$

$$p_v = p_0 \quad \text{for } -2d \leq z \leq 0$$

$$p_v = -p_0 \quad \text{for } 0 \leq z \leq 2d$$

$$p_v = 0 \quad \text{Elsewhere}$$

$$(b) \quad z=0, -a \leq x \leq a, -b \leq y \leq b$$

$$\Phi = \int_S \mathbf{D} \cdot d\mathbf{A}$$

$$z=0; \quad \mathbf{D} = \rho_0(0+2d)\hat{\mathbf{a}}_z = 2d\rho_0\hat{\mathbf{a}}_z$$

$$z=0; \quad \mathbf{D} = -\rho_0(0-2d)\hat{\mathbf{a}}_z = 2d\rho_0\hat{\mathbf{a}}_z$$

The flux through the surface $z=0$ is

$$\Phi = \int_{-a}^a \int_{-b}^b 2d\rho_0 dx dy$$

$$\Phi = 2d\rho_0(2a)(2b)$$

$$\Phi = 8abd\rho_0$$

$$(c) \quad -a \leq x \leq a, -b \leq y \leq b, -d \leq z \leq d$$

$$Q = \int_{-a}^a \int_{-b}^b \left(\int_{-d}^0 \rho_0 dz + \int_0^d -\rho_0 dz \right) dy dx$$

Calculate the integrals

$$-d \leq z \leq 0: \int_{-d}^0 \rho_0 dz = \rho_0 d$$

$$0 \leq z \leq d: \int_0^d -\rho_0 dz = -\rho_0 d$$

$$Q = \int_{-a}^a \int_{-b}^b (d\rho_0 - d\rho_0) dy dx = 0$$

(d) $-a \leq x \leq a$, $-b \leq y \leq b$, $0 \leq z \leq 2d$

$$Q = \int_{-a}^a \int_{-b}^b \int_0^{2d} -\rho_0 dz dy dx$$

$$Q = \int_{-a}^a \int_{-b}^b (-2d\rho_0) dy dx$$

$$Q = -2d\rho_0(2a)(2b)$$

$$Q = -8abd\rho$$

Q #05

$$D = 5.00 r^2 \hat{a}_r \text{ mC/m}^2 \quad r \leq 0.08 \text{ m and}$$

$$D = 0.205 \hat{a}_r / r^2 \text{ } \mu\text{C/m}^2 \text{ for } r \geq 0.08 \text{ m}$$

Solution:-

(a) ρ_v for $r = 0.06 \text{ m}$

$$\nabla D = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D_r)$$

$$\nabla D = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot 5r^2)$$

$$\nabla D = \frac{1}{r^2} \frac{\partial}{\partial r} (5r^4)$$

$$\nabla D = \frac{1}{r^2} (20r^3)$$

$$\nabla D = 20r$$

$$\nabla D = (20)(0.06)$$

$$\nabla D = 1.2 \text{ mC/m}^2$$

(b) p_v for $r = 0.1 \text{ m}$

$$\nabla D = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 D r)$$

$$\nabla D = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \cdot \left(\frac{0.205}{r^2} \right) \right)$$

$$\nabla D = \frac{1}{r^2} \frac{\partial}{\partial r} (0.205)$$

$$\nabla D = \frac{1}{r^2} (0)$$

$$\boxed{\nabla D = 0}$$

(c) $r = 0.08 \text{ m}$ to cause $D = 0$ for $r > 0.08 \text{ m}$

$$D_{\text{above}} - D_{\text{below}} = p_s$$

$$\frac{(0.205)}{(0.08)^2} - 5(0.08)^2 = p_s$$

$$p_s = \frac{0.205}{0.0064} - 5(0.0064)$$

$$p_s = 32.03 - 0.032$$

$$p_s = 32.03 - 32$$

$$\boxed{p_s = 0.03 \text{ } \mu\text{C/m}^2}$$